

Final Term

SQA

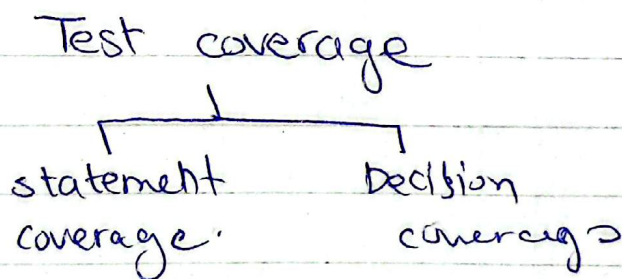
White box Testing (structure testing)
(Glass box testing)

↳ Focus on code structure
not on logic (output)

* Test coverage measurement

* Structural test case design

↳ Used for code optimization.



Statement coverage

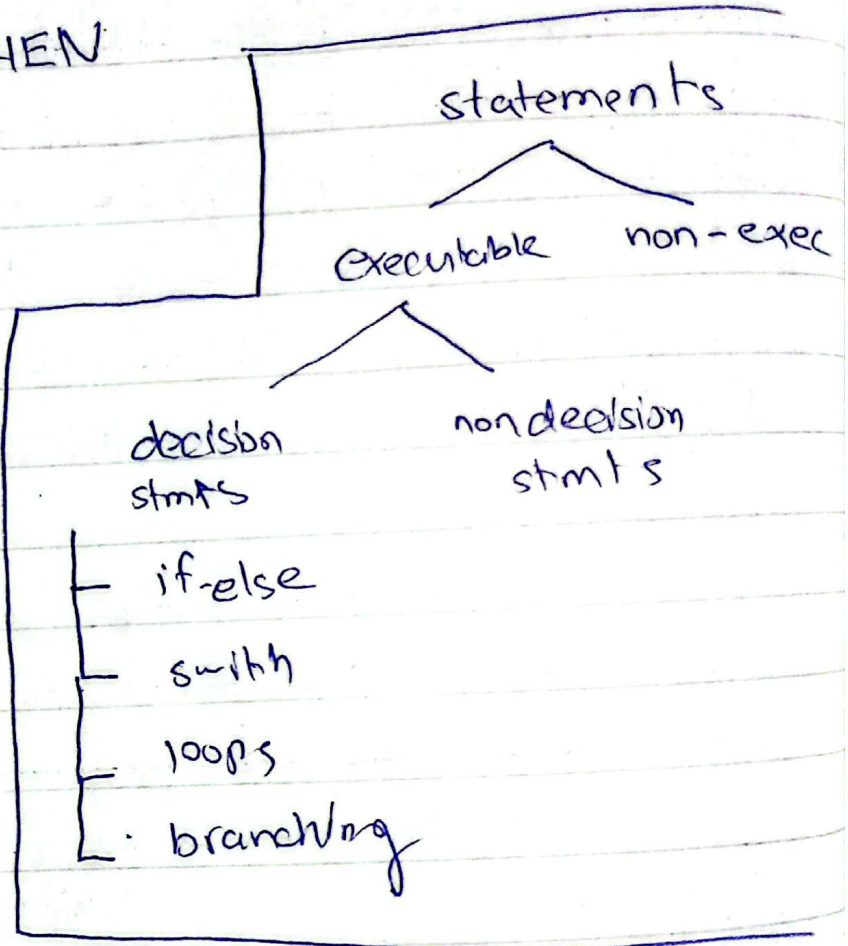
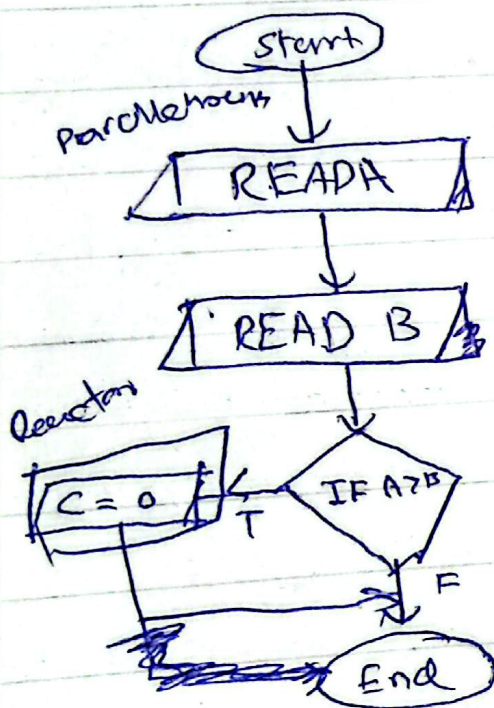
$$\text{statement coverage} = \frac{\text{no. of statements exercised}}{\text{Total number of stmts}} \times 100$$

↳ Objective: To cover ~~test~~ max (all) stmts.

code

```

Read A
READ B
IF A > B THEN
    C = 0
END IF
    
```



A	B	lines of stmts covered	no of stmts exercised	statement coverage %
1	2	1, 2, 3, 5	4	$4/5 \times 100 = 80\%$
2	1	1, 2, 3, 4, 5	5	$5/5 \times 100 = 100\%$

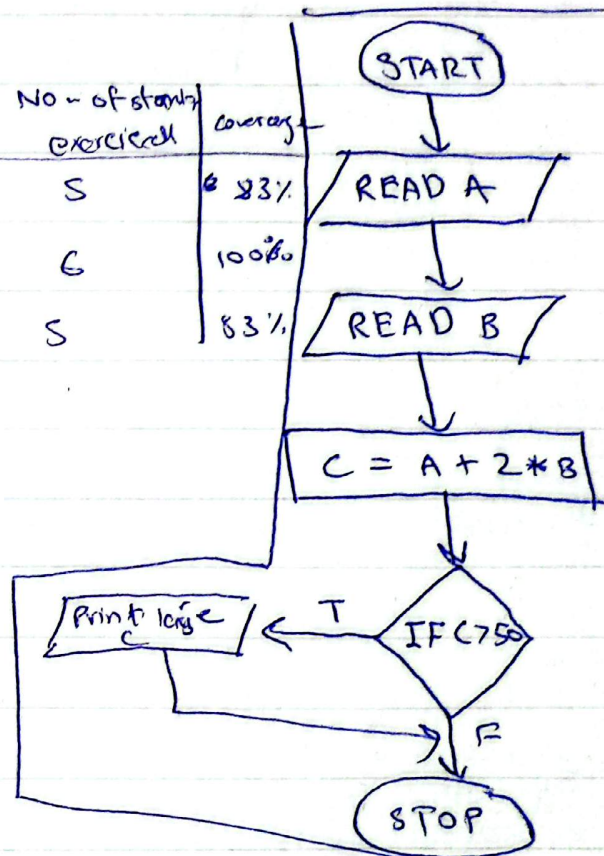
~~1~~ Only 1 test case req

example

```

1 READ A
2 READ B
3 C = A + 2 * B
4 IF C > 50 THEN
5     PRINT length C
6 ENDIF
    
```

A	B	C	lines of stmts covered	No. of stmts exercised	Coverage
10	20	50	1, 2, 3, 4, 6	5	83%
11	20	51	1, 2, 3, 4, 5, 6	6	100%
9	20	49	1, 2, 3, 4, 6	5	83%



example week 11

gold award	business class full?	economy full?	stmts covered	stmts exercise	coverage
Y	N	X	1,2,4,5	4	40 %
Y	Y	X	1,2,3,5	4	40 %
N	X	N	1,6,7,5	4	40 %
N	N	Y	1,6,8,9,5	5	50 %
N	Y	Y	1,6,8,10	4	40 %

Thursday

SQA

14/05/26

Decision coverage = $\frac{\text{No. of decisions exercised (path)}}{\text{Total number of decision outcomes (paths)}} \times 100$

Week 10
session 1

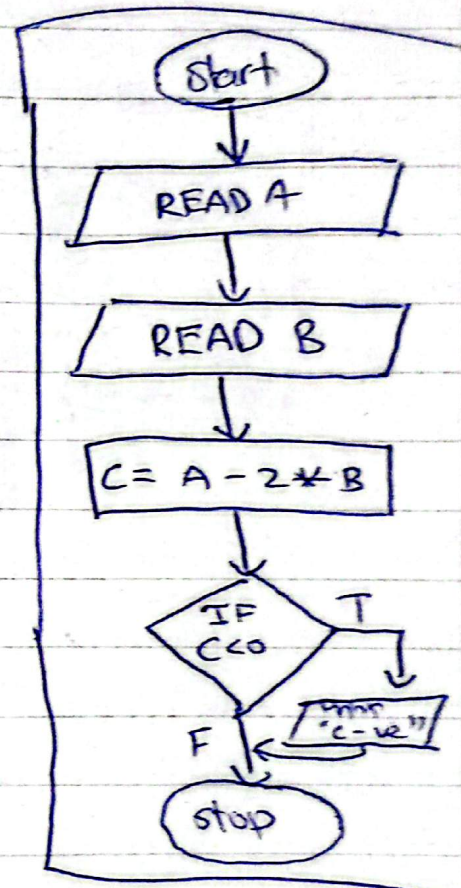
example

1. READ A
2. READ B
3. $C = A - 2 * B$
4. IF $C < 0$ THEN
5. PRINT "C negative"
6. END IF

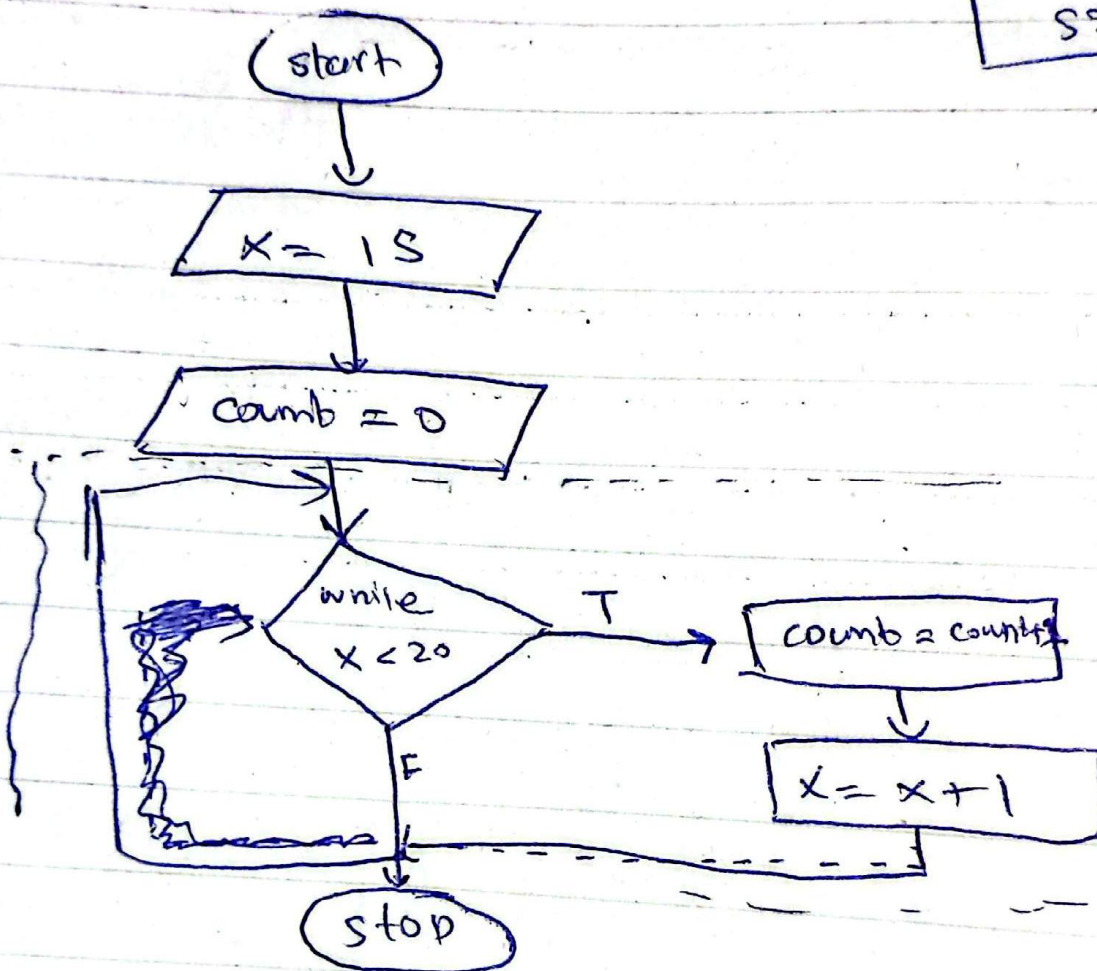
sol

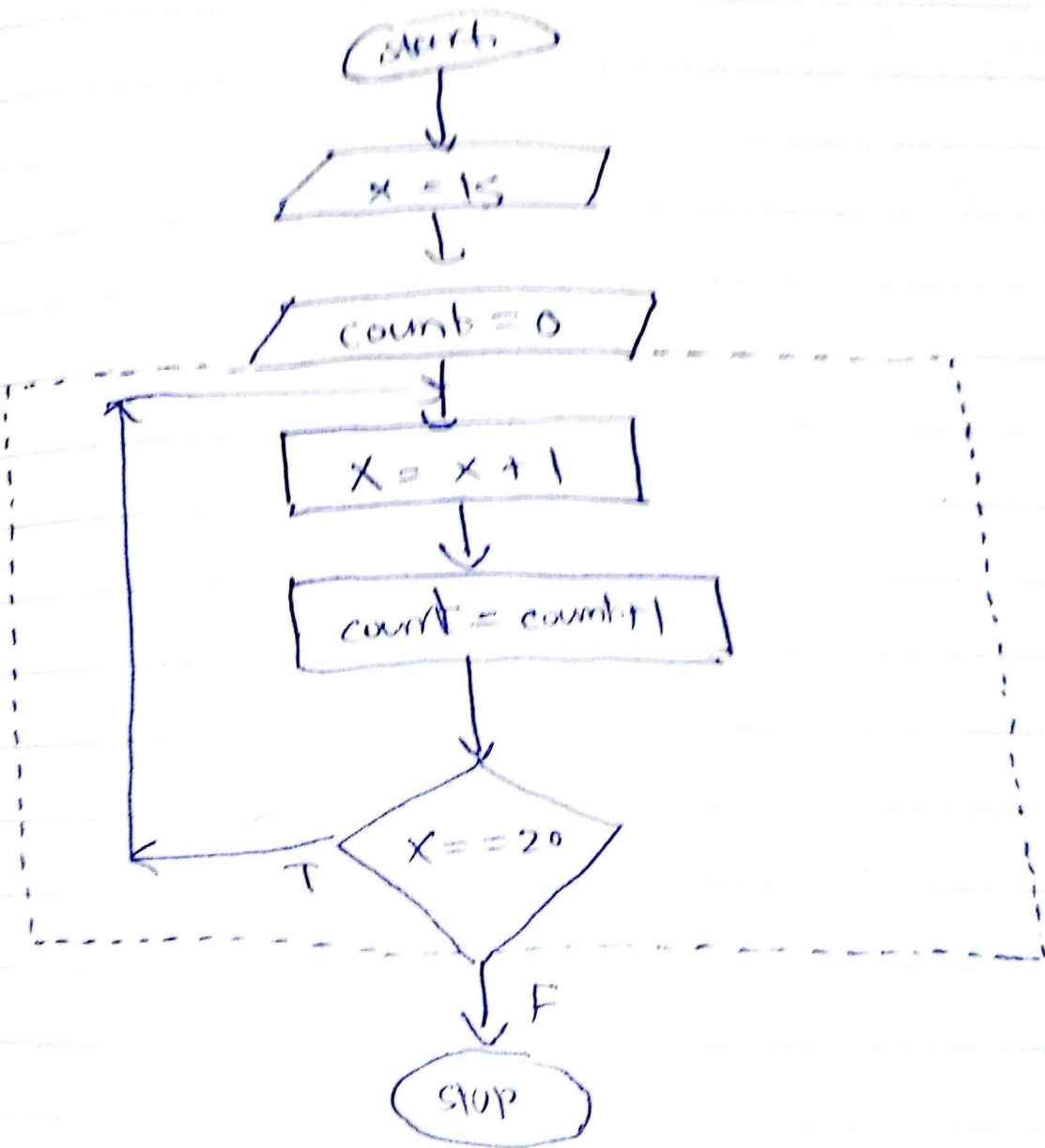
Total no. of decision outcomes = 2

A	B	C	line no. of stmts exec in path	Total no



pic in grd
memory





19/5/26

SQA

week 11 SL

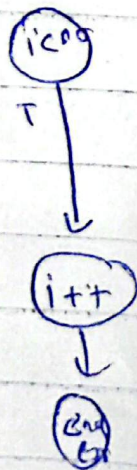
slide 4

* Control Flow Graph

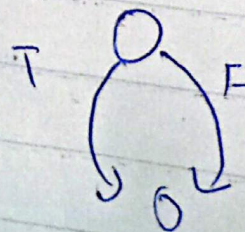
- used for finding decision coverage
- only considers decision statements

for($i=0$; $i < 10$; $i++$)

$i=0$



pic in
a
sad chat



True always
on left

slide 5

Step 1: Rewrite code by eliminating non executable and non decision statements.

```
13. IF A > B
14. THEN
15.     IF A > C
16.     THEN
18.     ELSE
20.     END IF
21 ELSE
22.     IF B > C
23.     THEN
25.     ELSE
27.     END IF
28. ENDIF
```

do
consider outermost if

13
19

21

28

consider THEN (true path)

15

16

18

20

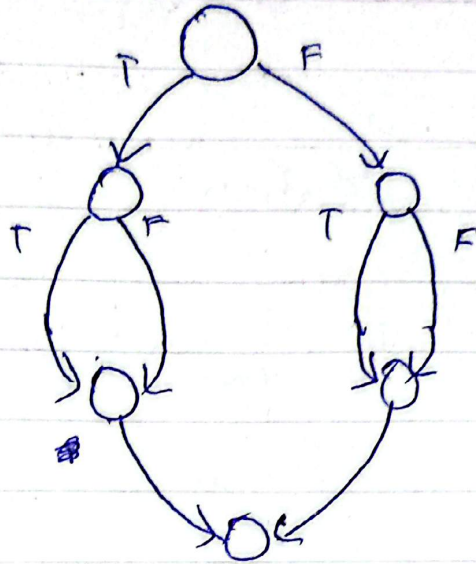
consider ELSE (false path)

22

23

25

27



Q1) How many ~~test~~ test cases

Pics in SQA
grp

wed

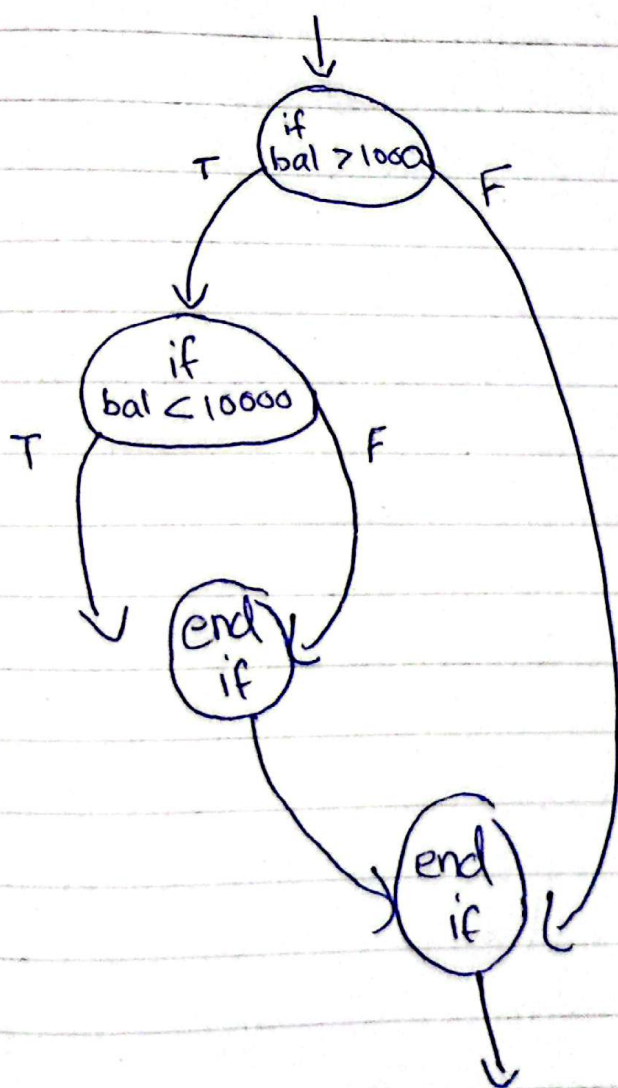
SQA

20/09/2026

HFG (Hybird Flow Graph)

step #1 Draw CFG:

week 11
SI



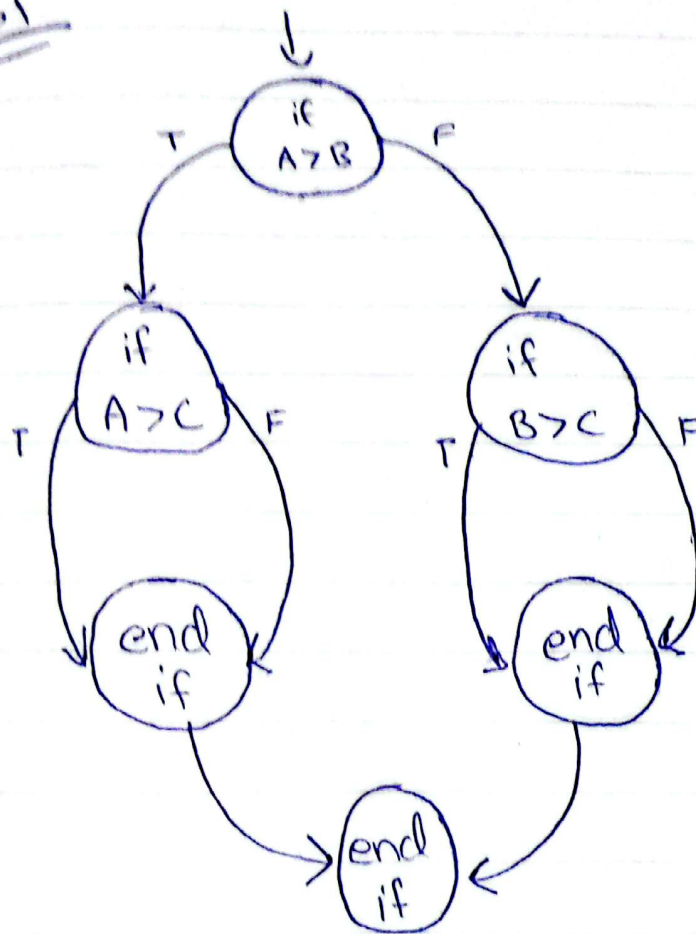
Step # 2

Rewrite the code by eliminating non executable stmts and grouping non decision stmts.

code

Step # 3 Draw HFG

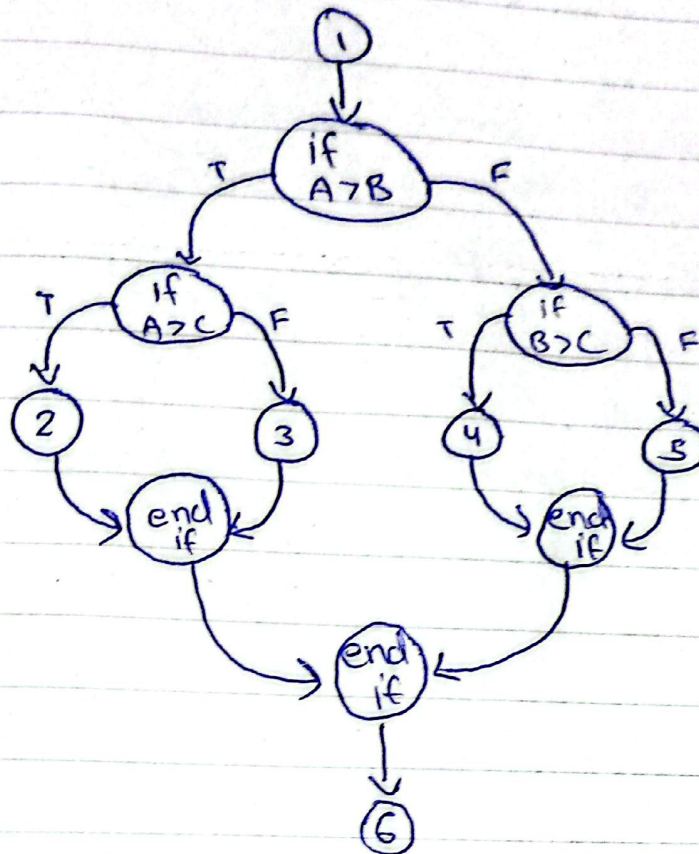
Example
Step 1



step 2

```
8. 1) Read A
9. 2) Read B
10. 3) Read C
11. 4) Mean = (A + B + C) / 3
13. 5) IF A > B
14. 6) THEN
15. 7)     IF A > C
16. 8)     THEN
17. 9)         Maximum = A      } ①
18. 10)     ELSE
19. 11)         Maximum = C     } ②
20. 12)     Endif
21. 13) ELSE
22. 14)     IF B > C
23. 15)     THEN
24. 16)         Maximum = B     } ③
25. 17)     ELSE
26. 18)         Maximum = C     } ④
27. 19)     ENDIF
28. 20) Endif
30. 21) Print ('Mean:', Mean)
31. 22) Print ('Max', Maximum) } ⑤
```

step 3 HFG



Paths

TT → A > B ✗ ✗ A > C → ① ② ⑥
 TF → A > B ✗ ✗ A <= C → ① ③ ⑥
 FT → A <= B ✗ ✗ B > C → ① ④ ⑥
 FF → A <= B ✗ ✗ B <= C → ① ⑤ ⑥

Time in SAT gcp

step 4

Q1) what level of stmt coverage would be achieved by the single input.

A=60 B=40 C=30

Sol

A	B	C	lines executed	no. of lines	stmt cov
60	40	30	1, 2, 3, 4, 5, 6, 7, 8, 9, 12, 20, 21, 22	13	$\frac{13}{27} \times 100$ = 51

Q2) How many test cases are required to achieve max stmt coverage

Sol 4

Q3) Generate test data for max stmt coverage

Sol

A	B	C
50	20	10
50	20	60
20	50	10
20	50	60

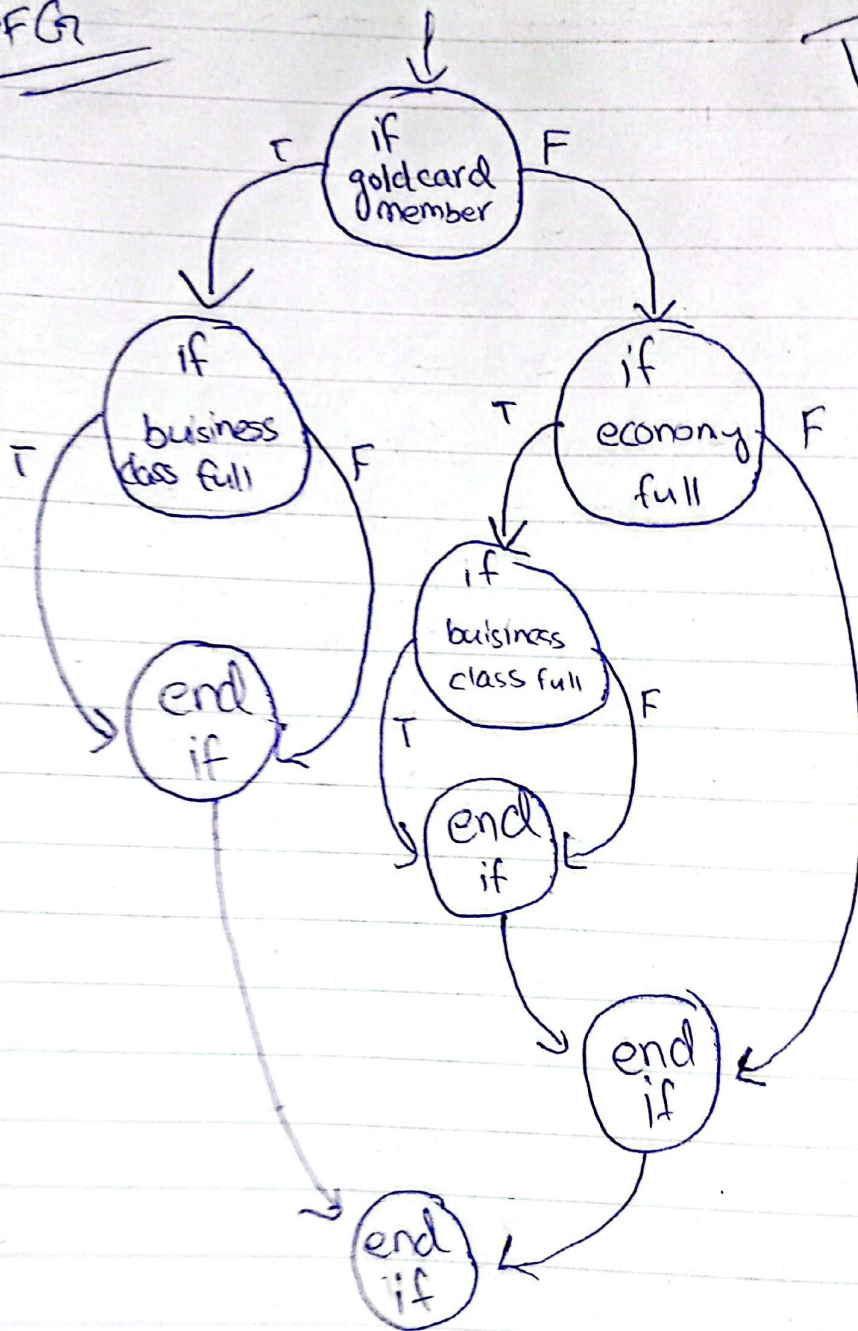
21/5/26

SQA

Thursday

CFG

code pic
in grp



Tests

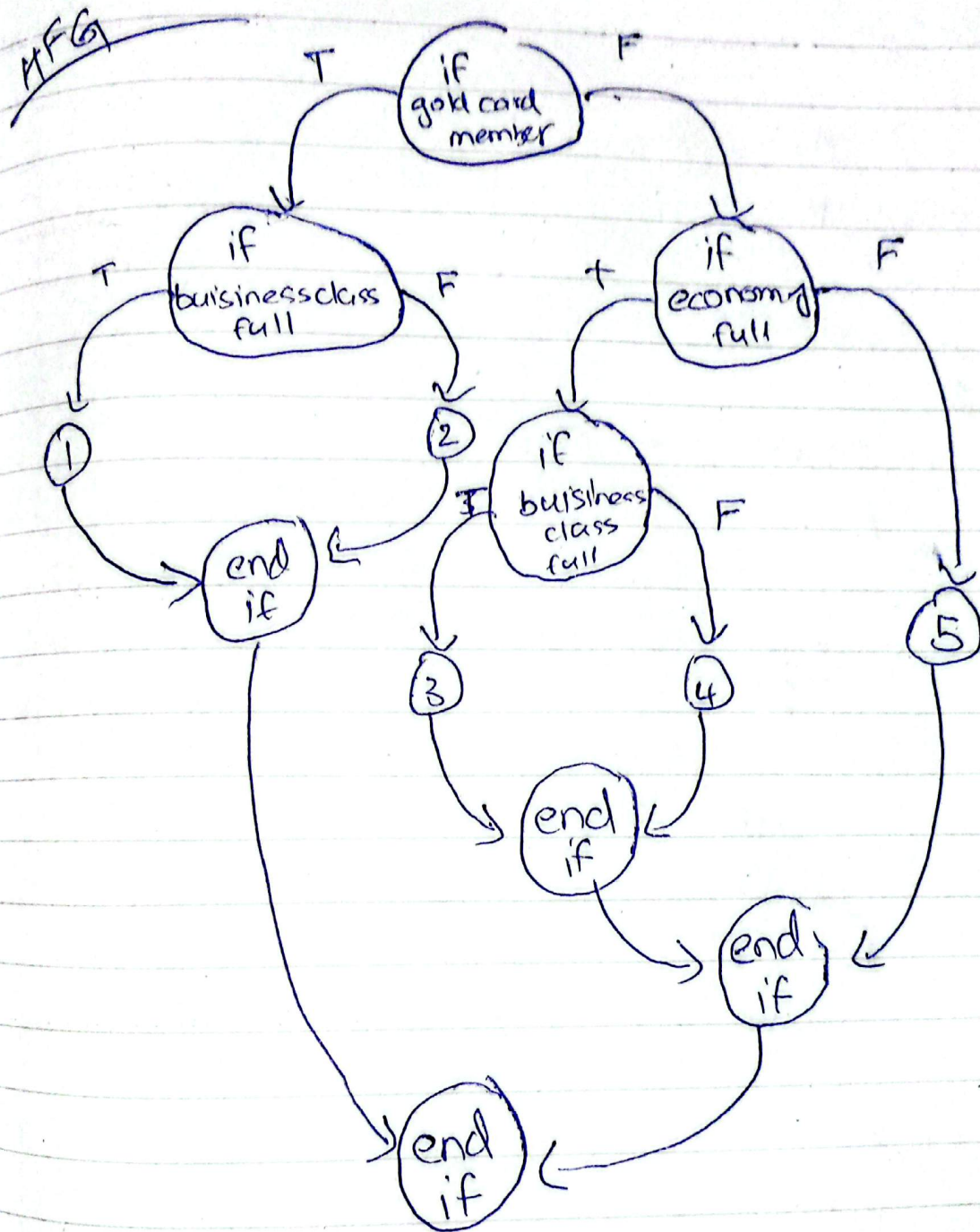
TT

TF

FTT

FTF

FF



Example

```
1 Read A
2 Read b
3 Read c
4 if (a > b) THEN
5     if
6     if (a > c) THEN
7         i = 0
8         while I < 10 DO
9             print a
10            END WHILE
11            print b
12            ELSE
13            print c
14            END IF
15        ELSE
16            print a, b, c
17        END IF
```

①

②

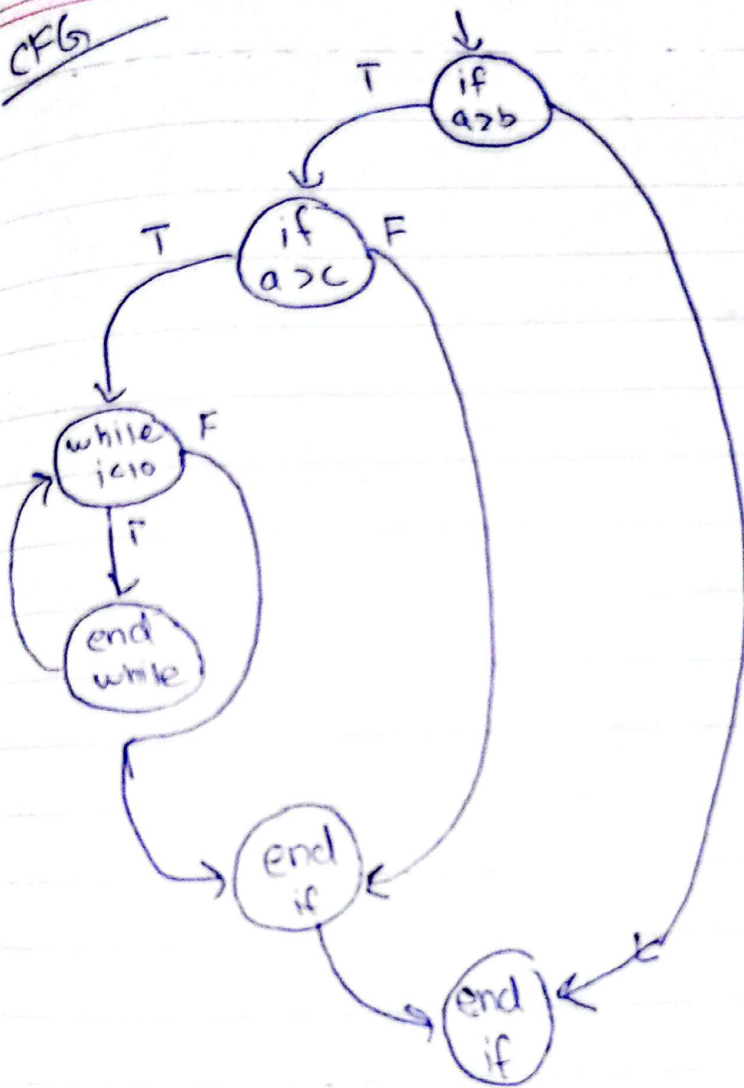
③

④

⑤

⑥

CFG



A	-22
B	-226
C	1106
X	1298
Y	902
Z	858

week 4

* Software ProcessProcess $\rightarrow$ steps / procedureSQA $\rightarrow$ Cost ($\downarrow$) & Quality ($\uparrow$)

$\hookrightarrow$ process improvement $\rightarrow$ continuous cycle

do 3

 $\downarrow$

20 marks

Quiz 3

last class
next week

Metric $\rightarrow$ the criteria for
standards

- $\hookrightarrow$ McCall $\rightarrow$ Product standard
- $\hookrightarrow$ CMM $\rightarrow$ Process standard
- $\hookrightarrow$ ISO $\rightarrow$ Both

* Capability Maturity Model $\hookrightarrow$ 5 levels

$\hookrightarrow$ By default every software dev company is level 1 and makes it level up one level at a time (incremental)

skip slide
5 and 6

* immature organisation

* mature organisation

* Levels

↳ Every level has associated KPA's (Key Process Areas)

↳ level 1

↳ immature organisation

↳ level 2

↳ Repeatable

Subcontract Management

↳ SLA's

wed

SQA

03/06/20

* CMM continue

→ level 3

→ level 4

week 4
slide 10

↳ Product improvement

↳ Process improvement

→ level 5 (optimizing level)

↳ 2 reasons

* output of level 4

* intro of new technology



* ISO software quality standards

↳ ISO 9126 (Product standard)

→ Quality characteristics → user perspective

↳ ISO 9000:2000 (SQA)

↳ ISO 9000

* fundamentals and vocab

* 8 principles

L) ISO 9001

* 5 kind of requirement

↳ ISO 9004

* Performance Improvement.

* ISO 9000

next
slide
23

ECO SMART TURBO

2H 4H 8H

-



+

Kurs

SQA

4/6/20

* ISO 9001

→ Remedial rev

↳ used for process and product improves

→ Realization Rev

↳ Rev gathering from customer

→ Resource Rev

→ Systematic Rev

↳ Top leadership

↳ Awareness

↳ create policy

→ Management Rev

↳ implementation system tree

clo 3	clo 2
week 4	week 10
week 13	week 11

week 13

* Incident Management

↳ unexpected events

* Bug tracking tool → Track all incident through closure

HW → Design a
incident report
template in excel

list steps to reproduce

Today

SQA

9-6-24

Week 13

↳ Configuration Management

- * Manages
 - ↳ source code
 - ↳ test scripts
 - ↳ 3rd party software
 - ↳ n/w
 - ↳ data
 - ↳ documentation

Approval

Name
Designation
~~Approval~~ Signature
Date

HW

pk in grp

Week 12 S1

Test organisation

↳ Test Independence

leave side 4 and 5

↳ Test approaches

- * Primitive (V-model)
- * Reactive (waterfall)

↳ Test strategies

↳ Test Planning

already covered
in
week 8 & 9

↳ baseline plan

↳ master test plan

↳ unit test plan

↳ integration test plan

↳ system test plan

↳ Acceptance test plan

↳ test plan in V-model

Hw
Design a
template for
test plan

↳ Test approaches

* Primitive (v-model)

* Reactive (waterfall)

↳ Test Strategies

↳ Test Planning

already covered
in
(weeks 8 & 9)

↳ baseline plan

↳ master test plan

↳ unit test plan

↳ integration test plan

↳ system test plan

↳ Acceptance test plan

↳ test plan in v model

Hw

Design a
template for
'test plan'

Wed

SQA

10/06/26

* Test estimation and planning :

Test estimation

week 12
S2

Metric
based

expert
based

↓
Depends on
Past similar
Project data

↓
Depends on
experience

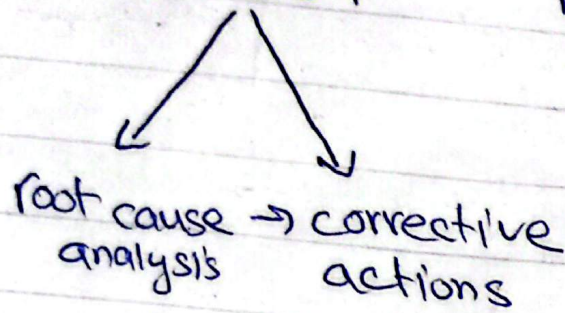
↓
Pre req: the data
must be accurate
for right estimation.

* Aspects used for estimation

- ↳ Product characteristics
- ↳
- ↳

* Test progress Monitoring

* Control



↳ To check if the test execution is according to the test plan if not then

* Subjective confidence of testers in the products.

repeat topic of week 5 S2

* Test reporting :

* exit criteria



Week 14

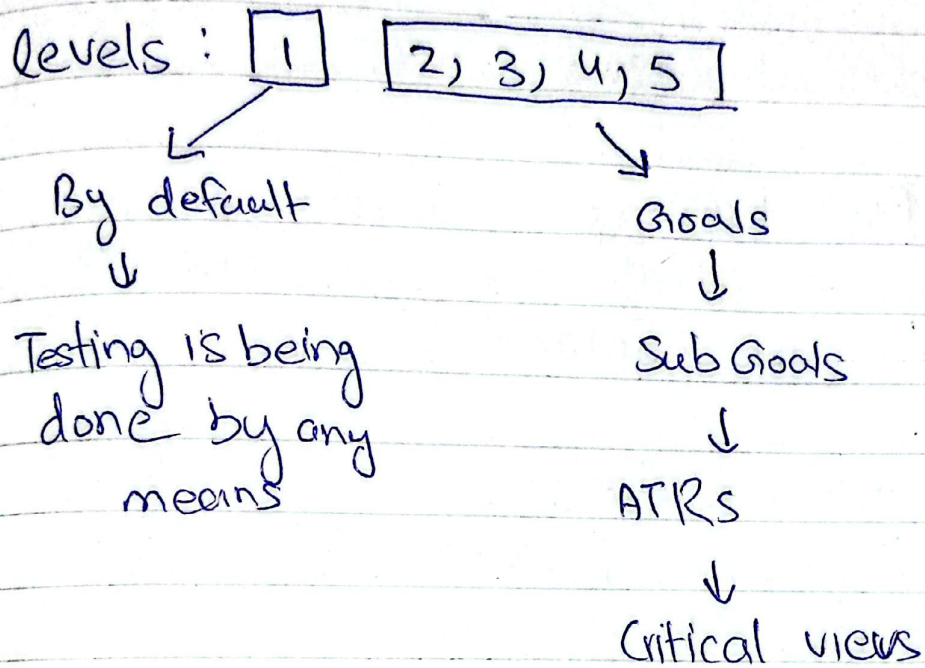
Testing Maturity Model (TMM)

HW

Design a template for 'Test Reporting' refer week 12 S2 slide 10 & 11

* Scope is testing processes only

(((



next class
TMM
level 3
slide 10

TMM - level 3

↳ Integration

↳ Independent test group

↳ Implement V model

↳ controlling and monitoring

TMM-level 4

↳

* CMM measured all processes while TMM measures only testing process

TMM level 5

Static testing

↳ Types

↳ Techniques